

Clase 3

$$\begin{aligned}x &= \int_{t_0}^{t_1} \left[u \ln \left(\frac{m_0}{m_0 - q t} \right) - g t \right] dt \\&= -g \int_{t_0}^{t_1} t dt + \int_{t_0}^{t_1} \ln[m_0] - \ln[m_0 - q t] dt \\&= \left. -\frac{g t^2}{2} \right|_{t_0}^{t_1} + \int_{t_0}^{t_1} \ln[m_0] dt - \int_{t_0}^{t_1} \ln[m_0 - q t] dt \\&= \left. -\frac{g t^2}{2} \right|_{t_0}^{t_1} + t \ln[m_0] \Big|_{t_0}^{t_1} - \int_{t_0}^{t_1} \ln[m_0 - q t] dt \quad \varphi = m_0 - q t \quad d\varphi = -q dt \Rightarrow dt = -\frac{1}{q} d\varphi \\&= \left. -\frac{g t^2}{2} \right|_{t_0}^{t_1} + t \ln[m_0] \Big|_{t_0}^{t_1} - \frac{1}{q} \int_{t_0}^{t_1} \ln \varphi d\varphi \\&= \left. -\frac{g t^2}{2} \right|_{t_0}^{t_1} + t \ln[m_0] \Big|_{t_0}^{t_1} - \frac{1}{q} \left[\varphi \ln \varphi - \varphi \right] \Big|_{\varphi_0}^{\varphi_1} \\&= \left. -\frac{g t^2}{2} \right|_{t_0}^{t_1} + t \ln[m_0] \Big|_{t_0}^{t_1} - \frac{1}{q} \left[(m_0 - q t) \ln(m_0 - q t) - (m_0 - q t) \right] \Big|_{t_0}^{t_1}\end{aligned}$$

Evaluando en: $u = 2000 \text{ m/s}$, $q = 2100 \text{ kg/s}$, $m_0 = 140\,000 \text{ kg}$, $g = 9.81 \text{ m/s}^2$, $t_0 = 8$, $t_1 = 30$.

$$x = 11\,057.16 \text{ m}$$